Prasad Sandip Joshi

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Education

Veermata Jijabai Technological Institute (VJTI) **Aug 2023 – May 2027**

Bachelor of Technology in Electronics and Telecommunication (Minor in DS & Machine Learning) *Mumbai, Maharashtra*

CGPA: 8.19/10 **MHT-CET:** 99.32 Percentile

Experience

Software Development Intern **June 2026**

Pidilite Industries Limited | Python, FastAPI, LangChain, Llama 3, Qdrant, PostgreSQL *Mumbai, Maharashtra*

- Contributed to **PilTantra+**, an enterprise **AI chatbot** enabling employees to retrieve accurate information from HR policies, SOPs, SharePoint resources, and internal knowledge bases using natural language.
- Developed and optimized a **Retrieval-Augmented Generation (RAG)** pipeline implementing document ingestion, intelligent chunking, vector indexing in **Qdrant**, semantic retrieval, and context-aware LLM response generation.
- Improved chatbot accuracy through **retrieval optimization**, **metadata filtering**, and **prompt engineering**, reducing hallucinations and improving answer relevance.
- Built and integrated **FastAPI** endpoints for document processing, retrieval, and chat interactions while collaborating cross-functionally to deliver a secure enterprise AI solution.

Relevant Coursework

- Database Management Systems
- Operating Systems
- Object Oriented Programming
- Data Structures & Algorithms
- Machine Learning
- Computer Networks

Projects

AI Financial Analyst | *Python, LangChain, Groq, FAISS, Pandas, RAG, Streamlit, Supabase* **June 2026**

- Developed an **agentic AI system** that autonomously analyzes financial CSV data and PDF annual reports through **natural language queries**, eliminating the need for manual formula writing or coding.
- Engineered a multi-tool **LangChain ReAct architecture** with autonomous tool selection across four components: a **Pandas agent** for code generation and execution, a **Plotly** chart generator, a FAISS-based **RAG** retrieval pipeline, and a direct LLM fallback.
- Built an end-to-end RAG pipeline using **PyPDFLoader**, **HuggingFace embeddings**,**OCR** support for scanned PDFs and **FAISS** for PDF question answering.
- Integrated **Supabase Authentication** with **persistent chat history** using **PostgreSQL**, enabling secure multi-user financial analysis.
- Deployed a Streamlit web application with **conversation memory**, **dynamic tool selection**, and interactive Plotly visualizations.

Credit Card Fraud Detection System | *Python, XGBoost, FastAPI, Streamlit, SHAP, Docker* **April 2026**

- Developed an end-to-end **fraud detection pipeline** using **SMOTE**, **PCA**, **Random Forest**, and **XGBoost** to classify highly imbalanced credit card transactions.
- Optimized the **XGBoost** model via **Optuna hyperparameter tuning** and threshold optimization, achieving **99.96% accuracy**, **89.32% precision**, **82.65% recall**, **85.86% F1-score**, and **98.26% ROC-AUC** on highly imbalanced transaction data.
- Built a **FastAPI REST API** and an interactive **Streamlit dashboard** supporting real-time and batch transaction predictions with **SHAP-based model** explainability.
- **Containerized** the application using **Docker** and implemented a basic **GitHub Actions CI/CD** pipeline to automate dependency installation, testing, and build verification while maintaining a modular and scalable project architecture.

NEP Timetable Management System | *Python, Genetic Algorithm, Node.js, JavaScript, MySQL* **October 2025**

- Designed and implemented a **Genetic Algorithm** to generate conflict-free academic timetables by optimizing faculty allocation, classroom assignment, and course scheduling under multiple constraints.
- Engineered the optimization pipeline using **selection**, **crossover**, **mutation**, and **fitness evaluation** to minimize scheduling conflicts and improve timetable quality.
- Developed a web-based management system with **role-based access** for administrators, faculty, and students using **Node.js**, **JavaScript**, and **MySQL**.
- Built a modular architecture separating optimization logic from the application layer for maintainability.

Technical Skills

Languages:	C++, Python, SQL
ML Libraries:	NumPy, Pandas, Scikit-learn, SHAP
Machine Learning:	Linear Regression, Logistic Regression, Decision Trees, Random Forest, XGBoost, PCA, SMOTE, Optuna
Frameworks:	FastAPI, Streamlit, LangChain
Databases:	MySQL, SQLite, NoSQL, PostgreSQL, Qdrant
Developer Tools:	Git, GitHub, Docker, VS Code

Leadership & Extracurricular

Enthusia, VJTI Mumbai **Aug 2023 – May 2026**

Volunteer

- Managed and coordinated the successful execution of the **sports** event as part of the college fest team.

Smart India Hackathon (SIH) 2025

Team Lead

- Led a team of four in developing an AI-based **NEP Timetable Management System**.